

ARKHANDIKA BUDI WIDODOPUTRA

Jakarta, Indonesia

+62 8xx-xxxx-xxxx

linkedin.com/in/arkhabw | github.com/arkhabw

PROFESSIONAL SUMMARY

Detail-oriented Software Developer specializing in backend architecture, native Android engineering, and full-stack web applications. Proven track record of developing interactive, algorithm-driven applications, designing efficient database schemas, and building secure REST APIs using Next.js, Kotlin, and Node.js. Passionate about writing clean, maintainable code, implementing intelligent AI algorithms (such as RAG and backtracking), and active contribution to student engineering groups.

TECHNICAL SKILLS

- Programming Languages:** Kotlin, Java, PHP, JavaScript, TypeScript, Python, HTML5, CSS3
- Web Development:** Next.js (App Router), React, Express.js, Tailwind CSS, Streamlit
- Mobile Development:** Android SDK, Jetpack Compose, Kotlin Coroutines, MVVM Architecture
- Database Systems & Cloud:** Database Design, MySQL, PostgreSQL, Firebase Cloud Firestore, Firebase Auth
- DevOps & Tooling:** Git, Docker, Postman, REST API Integration, RESTful APIs, Google OAuth
- AI & Optimization:** Retrieval-Augmented Generation (RAG), Backtracking Search (Branch-and-Bound), Dijkstra's Shortest Path, Fuzzy Logic, Bayesian Networks

EDUCATION

UPN "Veteran" Jakarta

Jakarta, Indonesia

Bachelor of Informatics (Computer Science)

2024 - Present

- Core Coursework:** Computer Science Fundamentals, Software Engineering, AI Development, Database Systems, Algorithmic Analysis
- Activities:** Active member in multimedia development groups and student organizations, maintaining a strong academic foundation.

EXPERIENCE

KSM Multimedia

Jakarta, Indonesia

Web Developer Member

2026 - Present

- Develop and deploy full-stack web applications, managing projects from initial UI/UX wireframes to backend logic and final deployment.
- Utilize HTML, CSS, React, and Next.js to construct high-fidelity, responsive, and accessible user interfaces.
- Design and manage database schemas (MySQL, PostgreSQL) and configure server logic using Node.js/Express to maintain secure and scalable systems.
- Collaborate with peers to conduct code reviews, ensuring adherence to coding guidelines and security best practices.

BEMF-IK UPNVJ

Jakarta, Indonesia

Human Resource Development Staff

2025 - 2026

- Supported the strategic management, evaluation, and professional development of BEMF-IK organizational members.
- Facilitated internal coordination, cross-divisional communications, and enforced organizational guidelines to maintain operational discipline.
- Contributed to designing and launching initiatives aimed at improving teamwork, member engagement, and overall cross-functional effectiveness.

PROJECTS

****RacePace AI - F1 Race Simulation & Pit Strategy Optimizer****

***Developer* | GitHub Link | Live Demo**

- Engineered a paddock telemetry race strategy simulator web app using Vanilla ES6 JavaScript, Chart.js, and Google Fonts.
- Developed an AI Strategy Engine using a backtracking search algorithm with branch-and-bound pruning to evaluate thousands of pit-lap and tyre compound combinations in milliseconds.
- Integrated complex race physics simulations including non-linear tyre wear ("cliffing"), fuel weight burn-off time effects (~0.06s/lap), and quadratic wet-track tyre grip scalings.

****Arnai - AI-Powered Study Companion****

***Developer* | GitHub Link | Live Demo**

- Built a comprehensive AI-powered study companion using Next.js (App Router), TypeScript, and Firebase.